soon as your play board is ready to use. It will increase by one second as we have set the interval property to '1000 milliseconds' in the designer for the clock component.

Now, when a tile is touched, how it should move is depicted in Figure 9. I have posted for two blocks. It will be the same for the others as well, keeping a separate event handler for all; so please create accordingly.

Application code

It was not possible to snap each and every block of the logic, so I am providing the complete code as a bar code. Scan the code (see QRCode 1) to obtain the *.aia* file, which you can upload to your projects in App Inventor 2 and view the complete code.

QRCode 1: Project source code file (.aia)

If you want to see how the application looks after installation, scan the code given in QRCode 2 to get the application *apk*.

QRCode 2: Application install File (.apk)

Figure 8: Block Editor image 5

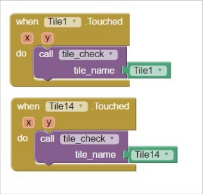

Figure 9: Block editor image 6

Packaging and testing

To test the app, you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the 'Build' button. It will show you the option to download the *apk* to your computer.
2. The progress of the download can be seen, and after it's been successfully done, the application will be placed in the *Download* folder of your directory or the preferred location you have set for it.
3. Now you need to get this *apk* file to your mobile phone either via Bluetooth or USB cable. Once you have placed the *apk* file to your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning saying,'Install from un-trusted source'. Allow this from the settings and after the successful installation, you will see the icon of your application in the menu of your mobile. Here, you will see the default icon, which can be changed and we will tell you how to do so as we move ahead in this course.

I hope your application is working exactly as per the requirements you have given. Now, depending upon your usage requirements and customisation, you can change various things like the image, sound and behaviour, too.

Debugging the application

We have just created the prototype of the application with very basic functionality, but what else might the user be interested in? Now come various use cases which require attention in order not to annoy the user. Your app should be able to address the following use cases:

1. Can we think of a way to share our moves and puzzle-solving time with our friends?
2. Can we display two boards simultaneously for two users and make it a dual player game?
3. What will other versions be like if we think of expanding the size of the square?

These are some of the scenarios that might occur and users will be pretty happy seeing these features implemented.

Think about all these scenarios and how you could integrate their solutions into the application. Do ask me if you fail to address any of the above cases.

You have successfully built another useful Android app for yourself. Happy inventing! END

By: Meghraj Singh Beniwal

The author is a B. Tech in electronics and communication, a freelance writer and an Android app developer. He is currently working as an automation engineer at Infosys, Pune. He can be contacted at *meghrajsingh01@rediffmail.com* or *meghrajwithandroid@gmail.com*